MONJUR E MOWLA

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CAREER OBJECTIVE

Seeking a position as Lecturer or Senior Lecturer in Computer Science and Engineering or a related department at a university, with an emphasis on practical teaching, student mentoring, and research-oriented projects.

EDUCATION

B.Sc. in Computer Science and Engineering | Ahsanullah University of Science and Technology

Passing year: 2023 | CGPA: 3.76/4.00 | Batch rank: 10th of 140

Thesis/final-year project: Blockchain-based Decentralized Election System

Higher Secondary Certificate (HSC) | Dhaka Imperial College

2015–2017 | GPA: 5.00/5.00

Secondary School Certificate (SSC) | Faizur Rahman Ideal Institute

2008–2015 | GPA: 5.00/5.00

PUBLICATION

“An Empirical Study of Pre-Trained CNN Models on COVID-19 CT Scan Images,” 4th International Conference on Sustainable Technologies for Industry 4.0 (STI), 2022.

PROFESSIONAL EXPERIENCE

Neural Semiconductor Limited | Uttara, Dhaka | Full-time, on-site | December 2022–Present

Senior Embedded Systems Engineer | January 2026–Present

- Developing edge-AI and embedded machine-learning workflows on the Radxa ROCK 3B single-board computer using the Rockchip NPU and RKNN-Toolkit2.

Embedded Systems Engineer III | January 2025–December 2025

- Worked on bare-metal STM32 programming, Zephyr RTOS, and MicroPython.

Embedded Systems Engineer II | January 2024–December 2024

- Worked with embedded Linux, Buildroot, the Linux kernel, Linux device drivers, and socket programming.

Embedded Systems Engineer I | March 2023–December 2023

- Built C++ virtual models using SystemC, TLM, and SCML for an LC3 (Little Computer 3), a counter, a DMA, and an LPUART.
- Connected the LPUART to a RISC-V core and developed its test bench.

Embedded Systems Engineer Trainee | December 2022–February 2023

- Worked on C/C++ programming and technical documentation.

TEACHING, MENTORING AND LEADERSHIP

Vice President, AUST Robotics Club | January–December 2022. Delivered workshops, trained team members, and helped plan events.

Judge, Project Showcasing Segment, AUST Rover Challenge 2.0 | July 2025. National robotics competition organized by AUST Robotics Club.

PROFESSIONAL AND ACADEMIC PROJECTS

Edge AI on Radxa ROCK 3B: Runs face-detection, face-alignment, and face-recognition models on the Rockchip NPU rather than the CPU, using RKNN-Toolkit2.

Voice-Controlled Automated Home (Arduino): Embedded system for controlling lights and fans from an Android application via Bluetooth.

JobFinder (ASP.NET MVC): Job platform where companies post vacancies and search candidates by skills; job seekers apply, upload CVs, write reviews, and receive recommendations.

NoWasteNoHunger (Android/Java): Application where users post leftover food for collection, with SMS alerts sent to the poster.

Covid Protection Store (PHP): E-commerce website for browsing and ordering products, with an admin panel for managing products and orders.

TECHNICAL SKILLS

Programming languages: C, C++, SystemC, Python, MicroPython, Java, C#, PHP, SQL, JavaScript, XML

Embedded platforms: STM32, Raspberry Pi, Radxa ROCK 3B, Arduino, QEMU

Embedded and systems programming: Firmware development, bare-metal programming, Zephyr RTOS, embedded Linux, Linux kernel and device drivers, Buildroot, socket programming

Modeling and architecture: SystemC, TLM, SCML, RISC-V, digital logic design, computer architecture

Edge AI and data: Rockchip NPU, RKNN-Toolkit2, PyTorch, NumPy, data structures and algorithms

Web, mobile and blockchain: ASP.NET, PHP, Android (Java), jQuery, Bootstrap, Three.js, Ethereum, Solidity, web3.js

RESEARCH AND PROFESSIONAL INTERESTS

Embedded systems and microcontroller interfacing; embedded Linux and real-time operating systems; IoT and edge AI; C/C++ and Python programming; digital logic design and computer architecture (including RISC-V); web and Android application development.

AWARDS, CERTIFICATIONS AND ACHIEVEMENTS

- Dean’s List of Honor, graduating batch of Fall 2021
- Gold-level Problem Solving badge, HackerRank
- Solved 300+ programming problems across Codeforces, HackerRank, UVa, HackerEarth, SPOJ, and CodeChef; participated in 50+ Codeforces contests (HackerRank: monjur_e_mowla; Codeforces: number27).
- HackerRank certificates: Problem Solving (Basic), December 2020; C# (Basic), September 2021; SQL (Intermediate), April 2022; Python (Basic).
- Coursera: Programming for Everybody – Getting Started with Python (June 2020); Python Data Structures (July 2020); Google Cloud Product Fundamentals (July 2020); Web Design for Everybody: Basics of Web Development and Coding Specialization (July 2020); Infrastructure and Application Modernization with Google Cloud.

REFERENCES

Dr. Mohammad Harun-Or-Rashid | Professor, Department of Mechanical and Production Engineering, Ahsanullah University of Science and Technology | harun_edu@yahoo.com

Mr. Hasib Md. Abid Bin Farid | Assistant Professor, Department of Electrical and Electronic Engineering, Ahsanullah University of Science and Technology | hasib.eee@aust.edu